CHAPTER 1: INTRODUCTION

1.1 Introduction

In modern industrial systems and electric mobility, real-time health monitoring and predictive maintenance of rotating machinery (such as bearings and motors) are critical for reducing downtime, minimizing maintenance costs, and ensuring operational safety. The **AURA-Twin** (*Automated Understanding and Real-time Analysis Twin*) project develops an intelligent Digital Twin ecosystem that goes beyond traditional binary status checking (Healthy/Faulty) towards a comprehensive diagnostic and decision-support framework.

By integrating physical sensor telemetry—specifically **vibration signals**—with multi-stage Artificial Intelligence models, the system continuously tracks mechanical behavior, detects unexpected deviations, classifies fine-grained fault types, and generates actionable maintenance recommendations in real time.

1.2 Scientific Problem

While deep learning models excel at supervised fault classification, real-world machinery frequently encounters novel operational conditions and unknown failure modes that were never included in the training dataset. Traditional supervised networks often misclassify these novel anomalies as known categories, leading to incorrect diagnostic alerts and unsafe operation.

Furthermore, existing diagnostic systems often stop at fault identification without providing automated, expert-level recommendations for immediate maintenance procedures, creating a gap between automated detection and human maintenance intervention.

1.3 Project Objectives

To resolve these challenges, the upgraded **AURA-Twin System for Graduation Project 1** introduces a three-tier intelligent processing pipeline:

1. **Unsupervised Novelty Detection (Isolation Forest):**

Processes raw vibration signals to flag unknown or unmapped anomalies (Novelty/Out-of-Distribution behavior) that require immediate human technical inspection.

2. **Supervised Fault Classification (Hybrid CNN-LSTM):**

If the anomaly aligns with known failure patterns, a hybrid Convolutional Neural Network and Long Short-Term Memory (**CNN-LSTM**) model classifies the exact fault type (e.g., Inner Race Fault, Outer Race Fault, Ball Fault).

3. **Engine Recommendation Engine & Interactive Dashboard:**

Maps identified fault types against an expert Knowledge/Rule Base to provide automated action protocols and risk levels, seamlessly presented on a real-time **Streamlit Dashboard** for monitoring personnel.

1.4 Summary of Previous Work (AURA-Twin Junior Project)

"In the preceding junior phase, the foundational architecture of the **AURA-Twin** (*Automated Understanding and Real-time Analysis Twin*) system was successfully established. The initial project focused on developing an end-to-end intelligent Digital Twin framework utilizing the **FordA dataset** to monitor vehicle engine health telemetry.

The core methodology relied on supervised deep learning models—specifically a hybrid **Convolutional Neural Network and Long Short-Term Memory (CNN-LSTM)** architecture—alongside traditional baseline models to perform binary classification (Healthy vs. Faulty engine status). The system successfully processed raw sensor signals, optimized training stability using advanced learning rate schedulers, and integrated a preliminary interactive **Streamlit Dashboard** to visualize real-time prediction outcomes and classification metrics."

1.5 Conclusion

This introductory chapter establishes the foundational paradigm of the upgraded **AURA-Twin** system as a vital cornerstone for the next generation of smart electric vehicles and industrial machinery. By transitioning from basic binary condition monitoring to a sophisticated multi-stage Digital Twin architecture, the project effectively addresses the key limitations of traditional diagnostic systems.

Through the synergistic integration of **unsupervised Isolation Forest algorithms** for early novelty detection, **hybrid CNN-LSTM networks** for fine-grained spatial-temporal fault classification, and an **expert Recommendation Engine** tied to a real-time **Streamlit Dashboard**, the proposed system ensures:

- **Enhanced Operational Safety:** By detecting both known mechanical degradation patterns and unknown out-of-distribution anomalies before critical failures occur.
- **Extended Asset Life:** By converting complex sensor telemetry into actionable maintenance protocols that prevent secondary component damage.
- **Real-Time Decision Support:** Providing monitoring personnel with transparent risk assessments and interactive virtual feedback.

In summary, the **AURA-Twin** project bridges the gap between raw physical vibration telemetry and intelligent maintenance intervention. The following chapters will detail the theoretical background, methodological architecture, and empirical implementation of this comprehensive Digital Twin framework.

CHAPTER 1: Basic Concepts

1.1 Introduction

To build a robust and intelligent Digital Twin for rotating machinery and electric vehicle condition monitoring, it is essential to understand the underlying technologies that drive the system's accuracy and reliability. This section provides a theoretical overview of the key components used in this project, ranging from the fundamental concept of the Digital Twin and vibration telemetry to advanced Deep Learning architectures—**CNN, LSTM, and Isolation Forest**. By integrating these methodologies, the system gains the ability to process complex vibration sensor data, recognize patterns, detect novel anomalies, and predict future states, creating a comprehensive digital shadow of the physical asset.

1.2 Basic Concepts & Core Technologies

1.2.1. Digital Twin Framework:

A Digital Twin is more than just a simulation; it is a dynamic virtual model that stays synchronized with the physical asset (such as electric motors and rolling element bearings) throughout its lifecycle. In this project, the Digital Twin uses real-time vibration telemetry and AI inference to mirror the degradation behavior of the mechanical components, providing a safe and transparent environment for performance analysis, anomaly detection, and fault prediction.

1.2.2. Vibration-Based Condition Monitoring (VBCM):

Vibration signal analysis is the cornerstone of rotating machinery diagnostics. Physical phenomena such as unbalance, misalignment, looseness, and bearing raceway defects generate characteristic high-frequency vibrations and modulation sidebands. Capturing and processing these raw vibration signals allows the system to identify mechanical degradation long before catastrophic failure occurs.

1.2.3. Unsupervised Novelty Detection (Isolation Forest):

While supervised models classify known states, real-world machinery frequently encounters unforeseen operational conditions. Isolation Forest is an unsupervised anomaly detection algorithm that isolates anomalies rather than profiling normal data points. By constructing an ensemble of isolation trees, anomalies—which are few and different—require fewer splits, making it highly efficient for detecting novel, out-of-distribution faults in real time.

1.2.4. Convolutional Neural Networks (CNN):

While traditionally used for image processing, 1D-CNNs are exceptionally effective for time-series and vibration signal analysis.

- **Role in Project:** The CNN layers act as spatial feature extractors. They scan the raw vibration telemetry to identify local patterns, peaks, and frequency-domain characteristics that signify mechanical stress or localized defects.
- **Advantage:** It reduces data noise and automatically extracts the most important hierarchical features without requiring manual feature engineering.

1.2.5. Long Short-Term Memory (LSTM):

LSTM is a specialized type of Recurrent Neural Network (RNN) designed to overcome the vanishing gradient problem, allowing it to retain information over long sequences.

- **Role in Project:** The LSTM layers handle the temporal aspect. Since mechanical wear and bearing degradation are cumulative progressive processes, LSTM tracks how these variables evolve over time.
- **Advantage:** It provides the sequential memory needed to predict future states based on historical vibration trends.

1.2.6. The Hybrid CNN-LSTM Architecture:

The Hybrid model is the core diagnostic engine of this project. It connects the CNN and LSTM in a sequential pipeline:

1. **Input:** Raw vibration data streams.
2. **CNN Phase:** Extracts complex spatial features and localized frequency patterns from the input windows.
3. **LSTM Phase:** Learns the sequential dependencies and temporal progression from the features extracted by the CNN.
4. **Output:** Accurate fine-grained fault classification (e.g., Inner Race, Outer Race, or Ball faults).

- **Why Hybrid?** This combination ensures the model understands both "what spatial anomalies are present right now" (CNN) and "how the degradation relates to previous operational cycles" (LSTM).

1.2.7. Expert Recommendation Engine & Streamlit Dashboard:

To bridge the gap between automated AI detection and human intervention, the system integrates an expert Rule/Knowledge Base that maps diagnosed fault types to immediate maintenance action protocols and risk levels. These insights are synthesized and rendered in real time through an interactive **Streamlit Dashboard**, providing monitoring personnel with a complete, user-friendly control center.

1.3 Conclusion

This chapter has established the theoretical and architectural foundations of the upgraded **AURA-Twin** system. By exploring the core principles of Digital Twins, vibration-based condition monitoring, unsupervised novelty detection, and advanced deep learning architectures, we have outlined the framework required for intelligent industrial prognostics.

The integration of **Isolation Forest** for unknown anomaly isolation, the **Hybrid CNN-LSTM** model for robust spatial-temporal fault classification, and the expert **Recommendation Engine** linked with the **Streamlit Dashboard** forms a cohesive, end-to-end ecosystem. This multi-tier approach successfully bridges the gap between raw physical telemetry and actionable maintenance insights, ensuring enhanced safety, operational reliability, and real-time decision support for modern rotating machinery. The subsequent chapters will detail the specific implementation methodologies and empirical evaluations of this framework.

CHAPTER 2: Study References

2.1 Introduction

The development of efficient monitoring systems and intelligent predictive maintenance (PdM) for rotating machinery and electric vehicles has been a focal point of recent research. Scholars have explored various methods ranging from traditional time-domain analysis to modern data-driven and digital twin approaches. This section reviews the evolution of condition monitoring, highlighting the shift toward vibration signal processing, unsupervised novelty detection, and advanced deep learning architectures.

2.2. Evolution of Rotating Machinery and Bearing Monitoring

Recent research has highlighted the critical role of advanced condition monitoring systems in ensuring the safety and operational efficiency of industrial machinery. Traditional methods often rely on simple threshold checks or statistical metrics, which struggle with non-linear dynamics and varying operational loads. The literature emphasizes that transitioning to comprehensive vibration-based condition monitoring (VBCM) is essential for capturing early-stage mechanical degradation such as unbalance, misalignment, and bearing raceway faults.

2.3. Deep Learning for Time-Series and Vibration Analysis

To address the limitations of conventional classifiers, data-driven approaches using Deep Learning (DL) have gained significant traction. Specifically, Convolutional Neural Networks and recurrent models have been widely adopted for feature extraction and pattern recognition in sequential sensor telemetry. Studies show that capturing temporal dependencies allows monitoring systems to track cumulative wear and progressive mechanical failure effectively.

2.4. Spatial Feature Extraction and Hybrid Models (CNN-LSTM)

The integration of Convolutional Neural Networks (CNN) with Long Short-Term Memory (LSTM) networks has introduced a powerful multi-dimensional perspective to vibration analysis. While LSTMs excel at tracking temporal degradation trends, 1D-CNNs are highly effective at extracting local spatial patterns and filtering noise from raw vibration streams. Recent literature emphasizes that hybrid **CNN-LSTM** models represent the state-of-the-art,

achieving superior accuracy (up to 96.00%) by combining spatial feature extraction with sequential memory, outperforming standalone architectures.

2.5. Unsupervised Novelty Detection (Isolation Forest)

A major challenge identified in modern machine learning diagnostics is the occurrence of unseen or out-of-distribution faults during real-world operation. Traditional supervised models often misclassify these unknown anomalies. Recent studies highlight the critical necessity of incorporating unsupervised algorithms—specifically **Isolation Forest**—as an intelligent first line of defense to isolate novel anomalies and unmapped operational conditions before deep classification is applied.

2.6. The Role of Digital Twins (DT) and Expert Recommendation Engines

Modern monitoring frameworks are increasingly shifting toward AI-empowered Digital Twins. A Digital Twin creates a dynamic virtual replica that works alongside physical telemetry, providing real-time data-driven insights without invasive hardware. Furthermore, advanced frameworks integrate expert Rule/Knowledge Bases to bridge the gap between automated AI detection and human intervention, translating diagnostic outputs into immediate, actionable maintenance recommendations.

2.7. Key Challenges in Industrial Condition Monitoring

According to reviewed literature (e.g., Kibrete et al., 2026; Parihar & Shivhare, 2026), several key challenges hinder effective predictive maintenance:

- **Non-linearity:** Mechanical behaviors and vibration signatures are highly non-linear, making traditional mathematical models inaccurate under fluctuating loads.
- **Sensor Noise & Artifacts:** Raw industrial sensor data often contains significant environmental and operational noise, which can mislead simple AI models.

- **Novelty and Data Scarcity:** Real-world machinery frequently encounters unforeseen fault states that are absent during training, necessitating unsupervised novelty detection layers.

2.8. Summary of Findings and Results

The studies provided in the reference documents demonstrate impressive empirical outcomes for advanced architectures:

- **High Diagnostic Accuracy:** Hybrid CNN-LSTM models demonstrated classification accuracies of up to 96.00% and F1-scores of 95.00% in vibration-based fault diagnosis tasks.
- **Superiority of Multi-Tier Pipelines:** Research proves that combining unsupervised anomaly detection (Isolation Forest) with supervised deep classification (CNN-LSTM) significantly reduces false alarm rates and enhances operational reliability.
- **Real-Time Decision Support:** The integration of AI-empowered Digital Twins with interactive dashboards enables continuous, proactive maintenance without requiring complex manual calibration.

Research Paper Summary:

1. Vibration-based condition monitoring and fault diagnosis of rotating machinery: Fundamentals, signal processing techniques, standards, and emerging trends

1. Introduction and Research Objectives

The study presents an extensive systematic review investigating the evolution, current state, and future directions of Vibration-Based Condition Monitoring (VBCM) and diagnostic systems for rotating machinery. The primary objective is to synthesize literature findings from 238 benchmark publications to establish a standardized framework connecting physical vibration phenomena, advanced signal processing, and modern industrial maintenance strategies.

2. AI/ML Tools and Signal Processing Techniques

The research examines a wide array of analytical and data-driven tools:

- **Vibration Analysis:** Categorizing mechanical excitation sources such as rotor unbalance, misalignment, looseness, and bearing raceway faults.
- **Advanced Signal Processing:** Reviewing time-domain statistical metrics, Fast Fourier Transform (FFT) spectral analysis, time-frequency transformations (Wavelet Transforms), and order tracking.
- **IIoT & Edge Integration:** Examining the transition from analytical models to data-driven deep learning and Industrial Internet of Things (IIoT) architectures.

3. Systematic Taxonomic Methodology

- Synthesizing multi-disciplinary literature to map the complete diagnostic pipeline from physical sensor placement to automated decision support.
- Aligning diagnostic health indicators with standardized ISO vibration severity criteria.

4. Key Results and Findings

- **Standardization of Diagnostics:** Established that combining rigorous vibration measurement with automated feature extraction significantly reduces false alarm rates in complex machinery.
- **Role of Digital Twins:** Highlighted that integrating real-time sensor telemetry with Digital Twin architectures bridges the gap between raw data acquisition and actionable maintenance prognostics.

2. Bearing Fault Diagnosis Using CNN-LSTM Hybrid Model For Predictive Maintenance.

1. **Introduction and Research Objectives**

The research discusses the development of an advanced deep learning framework for accurate, automated fault diagnosis in rotating machinery using rolling element bearing vibration data. The primary objective is to overcome the limitations of traditional machine learning classifiers by effectively capturing both spatial and temporal characteristics from raw vibration signals to achieve high diagnostic accuracy under varying operational conditions.

2. **AI/ML Tools and Techniques**

The study utilized specialized neural network layers for feature extraction and sequence modeling:

- **1D Convolutional Neural Networks (Conv1D):** Employed for automatic spatial feature extraction and frequency-domain pattern recognition directly from raw vibration telemetry.
- **Long Short-Term Memory (LSTM):** Integrated to process convolutional feature maps sequentially and capture long-term temporal dependencies and degradation trajectories.
- **Baseline Models:** Evaluated against traditional classifiers (such as KNN and Decision Trees) for performance benchmarking.

3. **Data-Driven Methodology**

- **Dataset:** Utilized the benchmark Case Western Reserve University (CWRU) bearing vibration dataset.
- **Data Splitting & Augmentation:** Partitioned data into 80% training and 20% testing subsets, enhanced with Gaussian noise injection to improve model generalization.

- **Optimization & Regularization:** Incorporated adaptive learning rate schedulers (ReduceLROnPlateau), dropout layers, and L2 regularization to ensure training stability and prevent overfitting.

4. **Key Results (AI Performance)**
    - **High Accuracy:** The hybrid CNN-LSTM model achieved an outstanding classification accuracy of **96.00%** and an F1-score of **95.00%**, significantly outperforming traditional baseline models (which hovered around 71%).
    - **Robustness:** Demonstrated high resilience against environmental noise and minor operational fluctuations in industrial vibration data.

3. Predictive Maintenance Using Machine Learning for Engineering Systems Through Real-Time Sensor Data and Anomaly Detection Models

1. Introduction and Research Objectives

The study focuses on developing an integrated Digital Twin framework that incorporates unsupervised learning algorithms to detect unforeseen anomalies and out-of-distribution faults in real time, addressing the limitations of purely supervised classification systems in industrial environments.

2. AI/ML Tools and Techniques

- Isolation Forest: Utilized as an unsupervised anomaly detection algorithm to isolate unknown or unmapped faults by constructing an ensemble of isolation trees.
- Supervised Classification Layers: Deployed downstream to categorize identified known failure modes (e.g., specific bearing faults).
- Expert Knowledge/Rule Base: Integrated to map diagnostic outputs directly to maintenance action protocols.

3. Data-Driven Methodology & Framework

- Multi-Tier Pipeline: Ingests raw vibration telemetry, routes data through the unsupervised novelty detection layer, and triggers supervised classification only when known operational states are confirmed.
- Dashboard Integration: Synthesizes diagnostic risk levels and action recommendations on a real-time interactive user interface.

4. Key Results and Adaptive Mechanisms

- Enhanced Reliability: Successfully mitigates the risk of misclassifying novel industrial anomalies, drastically lowering false alerts during shifting operational loads.
- Actionable Decision Support: Seamlessly bridges the gap between automated AI telemetry processing and human technical intervention, ensuring proactive maintenance across the asset's lifecycle.

4. Trustworthy Bearing Fault Diagnosis Based on Evidential Deep Learning and Quadratic Convolution Neural Network

1. Introduction and Research Objectives

The study focuses on developing a trustworthy and noise-robust bearing fault diagnosis framework that addresses the critical limitations of standard deep learning models—namely, their black-box decision processes and high susceptibility to overconfident misclassifications under severe industrial noise and uncertain operating environments.

2. AI/ML Tools and Techniques

- Quadratic Convolutional Neural Network (QCNN): Replaces traditional linear neurons with quadratic operations to enhance non-linear feature extraction and better capture complex transient impulse signals.

- Qttention Mechanism: A parameter-free attention module embedded directly via quadratic matrix factorization to adaptively focus on fault-relevant impact regions while filtering background noise.
- Evidential Deep Learning (EDL): Replaces standard Softmax layers by mapping outputs to Dirichlet distribution parameters, enabling direct quantification of predictive uncertainty mass.
- Improved Evidential Loss Function: Combines over-variance constraint penalty relaxation (R-EDL) with Fisher Information Matrix (FIM) dynamic weighting to eliminate overconfidence without over-penalizing high-uncertainty samples.

3. Data-Driven Methodology & Framework

- Noise-Resistant Feature Processing: Utilizes a wide $64 \times 1 \times 16$ kernel in the first layer to suppress high-frequency operational noise and extract core periodic fault patterns.
- Dual-Output Decision Pipeline: Generates a joint output comprising both the specific diagnostic fault classification and a trustworthy quantitative uncertainty score ($u = M/S$).
- Evidence-Based Risk Thresholding: Evaluates predictive confidence in real time, routing high-uncertainty samples ($u > 0.8$) to maintenance review rather than blindly trusting potentially flawed predictions.

4. Key Results and Adaptive Mechanisms

- High Robustness Under Severe Noise: Achieved $> 99.9\%$ accuracy in noise-free conditions and maintained superior diagnostic precision ($97.53\% - 99.91\%$) under extreme Gaussian noise (-6 dB on the CWRU dataset), outperforming baseline architectures like WDCNN, AResNet, and DCA-BiGRU.

- Reliable Uncertainty Separation: Successfully demonstrated a clear boundary where uncertainty scores rose above 80% for misclassifications and remained below 40% for correct predictions.
- Actionable Decision Support: Provides interpretability through visual Qttention maps and explicit uncertainty metrics, giving engineers a clear basis to schedule proactive maintenance safely.

5. Interpretable and Noise-Robust Bearing Fault Diagnosis for CNC Machine Tools via Adaptive Shapelet-Based Deep Learning Model

1. Introduction and Research Objectives

The study focuses on developing an interpretable and noise-robust bearing fault diagnosis framework for CNC machine tool spindles, addressing the critical limitations of traditional deep learning models—specifically their "black-box" nature and severe performance degradation under real-world industrial noise environments contaminated by cutting forces and cooling systems.

2. AI/ML Tools and Techniques

- Adaptive Multi-Scale Shapelet Extraction Module: Learns discriminative subsequence patterns directly connected to physical fault mechanisms to ensure inherent interpretability.
- Gated Parallel CNN: Employs depthwise separable convolutions across multiple scales for efficient spatial feature extraction and noise suppression via learnable gated weights.
- Residual-Enhanced BiLSTM: Models complex temporal dependencies and enhances discriminative capabilities under non-stationary noise conditions.
- Composite Loss Function: Combines cross-entropy, supervised contrastive learning, and multi-scale consistency regularization to optimize feature separability and model robustness.

3. Data-Driven Methodology & Framework

- Shapelet Matching Mechanism: Ingests raw vibration telemetry and matches local time-series patterns against learned shapelets, rendering the model naturally invariant to amplitude variations.
- Multi-Branch Feature Fusion Pipeline: Integrates spatial features from the gated parallel CNN with temporal representations from the residual BiLSTM.
- Robustness Validation Pipeline: Injects diverse real-world industrial noise profiles (Gaussian, uniform, and impulse noise) at varying Signal-to-Noise Ratios (SNR) to rigorously test performance limits.

4. Key Results and Adaptive Mechanisms

- Superior Noise Robustness: Achieved high diagnostic accuracy (e.g., 87.67% under -6 dB SNR Gaussian noise on CWRU) and maintained stable performance under severe industrial noise contamination, outperforming state-of-the-art baselines.

- Cross-Dataset Generalization: Validated successfully on both the benchmark CWRU dataset and the real-world IMS dataset, demonstrating strong generalization capabilities beyond controlled laboratory settings.

- Inherent Interpretability: Successfully bridges data-driven feature extraction with physical fault mechanisms, providing transparent decision-making support for predictive maintenance in CNC machining.

2.9 Conclusion

This research concludes that the AI-empowered Digital Twin approach significantly outperforms traditional condition monitoring methods for rotating machinery and electric mobility. By providing a multi-tiered diagnostic framework that combines unsupervised novelty detection with hybrid deep learning classification and real-time expert recommendations, this technology ensures early fault isolation, prevents catastrophic industrial failures, and contributes to the overall reliability and safety of smart industrial and transportation systems.

Paper Title	Year	Dataset Used	Training Approach	AI Model	Feature Extraction	Evaluation& Results
Vibration-based condition monitoring and fault diagnosis of rotating machinery: Fundamentals, signal processing techniques, standards, and emerging trends.	2026	Comprehensive Review Article; synthesizes data and findings from 238 peer-reviewed publications and international standards (e.g., ISO 10816, API 670).	Reviews Supervised Learning, Transfer Learning, Domain Adaptation, Unsupervised/Self-Supervised Learning, and Physics-Informed Hybrid Modeling.	Covers Traditional Machine Learning (SVM, Random Forest), Deep Learning (1D-CNN, LSTM, Autoencoders), and Explainable AI (XAI).	Time-domain (RMS, Kurtosis), Frequency-domain (FFT), Time-frequency domain (STFT, Wavelets), and adaptive signal decomposition (VMD, EMD, Envelope Analysis).	Demonstrates superior diagnostic accuracy using adaptive decomposition and hybrid deep networks for non-stationary signals. Highlights Digital Twins, Physics-Informed AI, and Edge AI as key future directions for predictive maintenance.
Bearing Fault Diagnosis Using CNN-LSTM Hybrid Model For Predictive Maintenance.	2026	Case Western Reserve University (CWRU) Bearing Dataset (incorporating Drive End (DE) and Fan End (FE) vibration signals).	Data Split: Stratified split of 70% training, 15% validation, and 15% testing. Data Augmentation: Gaussian noise injection, random amplitude scaling, and	Dual-branch Hybrid CNN-LSTM Architecture: Time-domain branch: Conv1D layers followed by an LSTM layer (64 units).	Automated End-to-End Feature Extraction combining Time-Domain raw vibration signals and Time-Frequency representations (STFT) without	Accuracy: 96.00% (Training accuracy reached ~97%) Precision: 95.00% Recall: 94.00% F1-Score: 95.00% Comparative Performance:

			random time-shifting. Optimization: Adam optimizer with Categorical Cross-Entropy loss over 20 epochs.	Frequency-domain branch: STFT spectrogram conversion followed by Conv2D layers. Feature fusion via Dense layers and classification with a Softmax output layer.	hand-crafted features.	Outperformed KNN (71.34%), Decision Tree (95.69%), and ResNet-50 + SVM (95.51%).
Predictive Maintenance Using Machine Learning for Engineering Systems Through Real-Time Sensor Data and Anomaly Detection Models	2024	The paper is a broad framework review and does not introduce a single primary dataset. However, it references standard time-series sensor data from IIoT environments (vibration, temperature, pressure, acoustics, current) used in various benchmark	The paper surveys multiple paradigms: Supervised Learning (for labeled historical fault data), Unsupervised Learning (for unlabeled anomaly detection), Semi-Supervised/Self-Supervised (Autoencoders), Hybrid/Ensemble	*Supervised:* Random Forest, XGBoost, SVM, Logistic Regression, KNN. *Unsupervised/Anomaly Detection:* Isolation Forest, K-Means, PCA. *Deep Learning:* Autoencoders, 1D CNN, LSTM, GRU, Hybrid CNN-LSTM.	Time-domain features (RMS, Kurtosis, Crest Factor), Frequency-domain features (Fast Fourier Transform - FFT), Spectral Entropy, and End-to-End automatic feature learning using 1D CNNs and LSTMs.	Discusses evaluation metrics such as Accuracy, F1-Score, AUC-ROC (for classification/anomaly detection) and RMSE (for Remaining Useful Life - RUL estimation). Summarizes that hybrid/deep models achieve high F1-scores and low

		case studies (e.g., CNC machines, jet engines, wind turbines).	Learning, and Federated Learning.			false-positive rates across manufacturing, rail, aviation, and energy deployments.
Trustworthy bearing fault diagnosis based on evidential deep learning and quadratic convolution neural network	2026	CWRU dataset (12 kHz drive-end bearing, 10 fault classes) contaminated with Gaussian white noise at 0 dB and -6 dB	Evidential Deep Learning (EDL) via Dirichlet distribution; improved loss function combining relaxed evidential loss, Fisher Information Matrix penalty, and KL divergence regularization; trained via SGD and ReLinear strategy	*EQCNN (Evidential Quadratic Convolutional Neural Network built on Wide Deep CNN architecture)*	Wide quadratic convolutional kernel ($64 \times 1 \times 16$) in Layer 1, embedded attention mechanism (Qttention), BN, ReLU, and Max Pooling across 6 QConv blocks	- Accuracy: 97.53%–99.91% at -6 dB SNR across motor loads, and 100% at 0 dB[- Uncertainty Quantification: Over 80% uncertainty for misclassifications and under 40% for correct predictions.Interpretability: Qttention maps accurately locate fault impact pulses
Interpretable and Noise-Robust Bearing Fault Diagnosis for CNC Machine Tools via	2026	CWRU (10 classes) and IMS run-to-failure datasets (4 classes) with Gaussian,	Composite loss function ($L_{CE} + \lambda_1 L_{cont} + \lambda_2 L_{cons}$)	*Adaptive Shapelet-based CNN-BiLSTM*	Multi-scale shapelets (lengths 16–256) with correlation-based distance,	- Accuracy: 87.67% (Gaussian), 85.65% (Uniform), and 82.33% (Impulse) at -

Adaptive Shapelet-Based Deep Learning Model		Uniform, and Impulse noise (-6 dB to +8 dB)	combining cross-entropy, supervised contrastive loss ($\lambda_1=0.05$), and multi-scale consistency regularization ($\lambda_2=0.02$) with K-means++ initialization		Gated parallel CNN (kernels 3, 5, 7), and Residual-enhanced BiLSTM	6 dB on CWRU; 99.58% on IMS (clean) Superiority: Statistically significant improvement ($p < 0.01$) over standard baselines .Interpretability: FFT spectrum aligns with theoretical fault frequencies (e.g., BPFO = 107.4 Hz)

CHAPTER 3: METHODOLOGY

3.1 Introduction

The methodology of this research follows a structured data-driven approach to develop a robust Digital Twin for vehicle health monitoring and industrial machinery diagnostics. The primary objective is to transform raw sensor telemetry into actionable diagnostic insights and proactive maintenance recommendations. This section outlines the comprehensive workflow, starting from data preprocessing and window segmentation to the implementation of a multi-tier Artificial Intelligence pipeline.

By integrating unsupervised anomaly detection with advanced deep learning architectures, the system achieves a hybrid balance between identifying unseen operational behavior and accurately classifying fine-grained mechanical failures. Convolutional Neural Networks (CNN) are employed for spatial feature extraction, while Long Short-Term Memory (LSTM) units handle temporal sequence modeling. A significant focus is placed on the development of the **Hybrid CNN-LSTM** model, which combines spatial feature extraction with sequential memory to enable high-precision, real-time predictive maintenance.

3.2 Dataset Description: The CWRU Bearing Dataset

To evaluate the upgraded AURA-Twin framework for multi-class vibration diagnostics, this project utilizes the **Case Western Reserve University (CWRU) Bearing Dataset**, a world-renowned benchmark for vibration-based condition monitoring (VBCM) and machine health prognostics.

- **Data Characteristics:** The dataset comprises high-frequency vibration acceleration signals gathered using accelerometers attached to the motor housing (at the drive end and fan end). The signals are collected under varying operational motor loads (0 to 3 horsepower) and motor speeds (1720 to 1797 RPM).
- Task: Unlike the binary acoustic classification (FordA) of the junior project phase, the task in Graduation Project 1 is a Fine-Grained Multi-Class Classification. The system distinguishes between normal operating conditions and localized mechanical degradation across different

components and severity sizes (e.g., Inner Race Faults, Outer Race Faults, and Ball Element Faults at $0.007''$, $0.014''$, and $0.021''$ fault diameters).

- **Significance for Digital Twin:** By processing raw vibration telemetry from the CWRU benchmark, the Digital Twin accurately simulates real-time mechanical degradation and dynamic operational stress in rotating components, serving as an intelligent virtual replica for predictive maintenance decision support.

3.3 Data Preprocessing and Signal Preparation

To ensure the deep learning models achieve optimal convergence and generalization, rigorous preprocessing steps are applied:

1. **Normalization & Scaling:** The implementation utilizes Z-Score Normalization (Standardization) to rescale vibration amplitudes:

$$z = \frac{x - \mu}{\sigma}$$

where x is the raw sensor reading, μ is the mean, and σ is the standard deviation. This stabilizes gradient descent during training.

2. **Window Segmentation & Tensor Shaping:** Continuous vibration time-series are segmented into fixed-length sliding windows. To ensure compatibility with TensorFlow/Keras 1D convolutional layers, tensor dimensions are reshaped to standard multi-dimensional arrays.

3. **Data Splitting:** The dataset is partitioned into 80% for training and 20% for testing to rigorously validate model generalization on unseen operational data.

3.4 Proposed System Architecture & Core Models

3.4.1 Unsupervised Novelty Detection (Isolation Forest)

Serving as the intelligent first line of defense, the Isolation Forest algorithm processes incoming vibration windows. By constructing an ensemble of isolation trees, it isolates out-of-distribution or unknown operational states

(novelties) that were absent during training, flagging them for immediate human technical review.

3.4.2 Supervised Fault Diagnosis: The Hybrid CNN-LSTM Model

When data corresponds to known operational states, it is routed to the core diagnostic engine:

1. **Convolutional Layers (Conv1D):** Automatically extract localized spatial features, peaks, and frequency-domain characteristics from raw vibration windows.
2. **Long Short-Term Memory (LSTM) Layers:** Process the spatial feature maps sequentially to capture long-term temporal dependencies and degradation trajectories.
3. **Regularization & Optimization:** Incorporating Dropout layers, L2 regularization, and adaptive learning rate schedulers (ReduceLROnPlateau) to prevent overfitting.

3.4.3 Expert Recommendation Engine & Streamlit Dashboard

The diagnosed fault outputs are mapped against an expert Rule/Knowledge Base to generate immediate maintenance action protocols and risk levels, presented seamlessly in real time through an interactive **Streamlit Dashboard**.